

Fundamentals of Enterprise Resource Planning

NISHAT TEXTILES



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Section 1 – Company Introduction

1.1 Company background

This report presents an enterprise resource planning (ERP) solution design and business analytics proposal for Nishat Textiles, treated in this project as a vertically integrated textile manufacturing and omni-channel retail organization. The case organization performs activities across the textile value chain, including raw material sourcing, spinning, weaving, dyeing, stitching, warehousing, retail selling and online order fulfilment. Because these activities are operationally connected, the central management problem is not only departmental efficiency but the flow of accurate information across departments.

In a textile business of this type, a sales order can affect several functions at the same time. It may reduce finished goods inventory, trigger replenishment, influence production planning, create a customer invoice, affect cash flow, and change the demand forecast for a future production run. If those functions use disconnected systems, the organization cannot maintain a reliable single view of products, stock, suppliers, customers, orders and labour. ERP is therefore appropriate because it is designed to integrate major business functions on a shared data and process foundation (Umble, Haft and Umble, 2003).

1.2 Industry type

The organization operates in textile manufacturing and omni-channel retail. Manufacturing creates complexity because production involves multiple stages, work centres, bills of materials, labour schedules and material movements. Retail creates a second layer of complexity because products must be visible and available across physical outlets and the online storefront. The proposed ERP design therefore needs to support both manufacturing control and customer facing fulfilment.

1.3 Existing systems and operations

The current case assumes a heavily customized legacy IT environment in which core manufacturing and warehouse transactions are maintained in an Oracle 10g database, while retail and ecommerce activities run through separate modern systems such as Shopify. Oracle Database 10g is an older technology generation; Oracle lifetime support documentation indicates that Oracle Database 10.2 reached general availability in 2005 and is outside the normal Premier and Extended Support windows (Oracle, 2025). This matters because unsupported legacy environments create long term risks around upgrades, security patching, integration, and availability.

The main operational weakness is fragmentation. Manufacturing, inventory, sales, finance and workforce data are not being managed as one continuous transaction chain. Online orders require manual or batch handling before they can be reconciled with warehouse records. Finance reports depend on delayed extracts and spreadsheets. Warehouse staff do not receive fully digital picking tasks now of order confirmation. The ERP proposal in this report addresses that fragmentation by creating one integrated operating model.

1.4 Project assumptions

The report is based on the academic case scenario rather than audited internal company records. The following assumptions are used transparently:

Assumption area	Explanation
Case organization	Nishat Textiles is treated as the organization described in the project scenario.
Current technology	The legacy environment is assumed to include Oracle Database 10g and a separate Shopify based online storefront.
Process timing	The current 24 hour manual cycle and proposed five-minute target cycle are treated as design targets, not audited measurements.
Integration model	Shopify and Odoo integration is treated as an API or connector implementation that requires testing, monitoring and exception handling.

This assumption section is included to make the analysis precise and academically transparent. It prevents the report from presenting design estimates as verified corporate performance data.

Section 2 – Problems in Current System

The current system problems are not isolated technical issues; they are linked business process failures. The key problems are duplicate data, inventory inaccuracy, weak integration, delayed reporting and poor cross-department communication. Each problem reduces the organization's ability to respond quickly to demand changes and to control operational cost.

Problem	How it appears in the case	Business impact
Duplicate data entry	Online orders and warehouse transactions must be entered or reconciled more than once.	Higher human error, duplicated customer/order records and time spent on rework.
Inventory issues	Physical stock, warehouse records and online availability are not synchronized continuously.	Overselling, customer cancellations, refunds and unreliable replenishment planning.
Lack of integration	Sales, inventory, manufacturing, finance and HR data exist in separate systems.	No single source of operational truth and weak traceability from order to fulfilment.
Delayed reports	Reports depend on manual extracts and spreadsheet consolidation.	Management sees problems after they have already affected customers or production.
Poor communication	Departments rely on manual handoffs, emails and exception lists.	Slow response time, unclear accountability and increased operational friction.

The most serious weakness is transactional fragmentation. A customer order begins in the online storefront, while stock validation and warehouse fulfilment are handled in a separate legacy environment. Because the systems do not operate as one workflow, staff must bridge the gap manually. This increases latency and creates a risk that different departments act on different versions of the same transaction. Shopify supports APIs and webhooks that can communicate order and inventory events to external systems, while Odoo provides external API capabilities for integration with enterprise data models (Shopify, n.d.-a; Shopify, n.d.-b; Odoo, n.d.-i). The problem is therefore not that integration is technically impossible; the problem is that the current architecture does not use integration in a controlled ERP workflow.

Inventory inaccuracy is also commercially damaging. In retail and e-commerce, a stockout is not only a warehouse event. It affects customer confidence, order fulfilment rate, refund workload and future demand. Liberopoulos and Tsikis (2019) show that stockouts can reduce both order fill rates and future customer demand. For Nishat Textiles, stockouts are especially risky during seasonal launches, where high online demand may occur before warehouse records are synchronized.

Delayed reports reduce decision quality. If managers wait for end-of-day exports or manual Excel consolidation, they cannot respond to a fast-selling product, a supplier delay or a production bottleneck in time. ERP data becomes valuable when it supports reporting, analysis and forecasting rather than simply storing completed transactions (Sastry and Prasada Babu, 2013).

Section 3 – ERP Solution Design

3.1 ERP software selection

The recommended ERP platform for this case is Odoo. Odoo is suitable because it provides modular business applications for inventory, manufacturing, purchase, accounting, sales, CRM, HR, attendances and payroll, while also offering external API capabilities for integration (Odoo, n.d.-a; Odoo, n.d.-b; Odoo, n.d.-c; Odoo, n.d.-d; Odoo, n.d.-e; Odoo, n.d.-i). The proposal does not claim that Odoo is universally better than every ERP suite. It argues that Odoo is a practical fit for the assignment scenario because the project needs an integrated, modular and adaptable ERP design rather than a single departmental system.

3.2 Reason for selecting Odoo

Selection factor	Explanation	Strategic value
Modular architecture	The company can implement Inventory, Sales, Procurement, Manufacturing, Finance and HR in phases.	Reduces go live risk and avoids a disruptive all-at-once migration.
Manufacturing fit	Odoo supports bills of materials, work centres, manufacturing orders and production analysis.	Matches the textile value chain from raw material to finished garments.
Inventory and warehouse control	Odoo Inventory supports stock moves, routes, lead times and replenishment rules.	Directly addresses overselling, stockouts and warehouse visibility problems.
Integration capability	Odoo external APIs and Shopify APIs/webhooks	Creates a realistic path for online order

Selection factor	Explanation	Strategic value
	allow connector-based integration.	synchronization.
Analytics support	ERP transactions can feed dashboards for sales, inventory, procurement, finance and HR.	Improves decision making and reduces reliance on manual Excel reporting.

A critical point is that Shopify integration should be treated as a managed integration project, not as a risk-free switch on feature. Odoo documentation lists specific marketplace connectors such as Amazon and Shopee in its sales documentation, while broader integration with external software is handled through APIs (Odoo, n.d.-e; Odoo, n.d.-i). Shopify likewise provides APIs and webhooks for store events (Shopify, n.d.-a; Shopify, n.d.-b). This means the proposed Odoo Shopify integration is feasible, but it must include middleware monitoring, retry logic, data validation and exception handling.

3.3 Expected improvements

The expected improvements are operational and managerial. Operationally, the ERP should reduce duplicate data entry, improve stock accuracy, automate replenishment triggers, support barcode based fulfilment and link customer orders to finance postings. Managerially, it should shorten reporting cycles, create reliable KPIs and make demand, production, procurement and labour information visible in one place. However, these improvements are expected outcomes, not guaranteed results. ERP success depends on disciplined implementation, business process redesign, training and governance (Umble, Haft and Umble, 2003; Garg, 2010).

Section 4 – ERP Modules

The ERP design uses six core modules required by the project: HRM, CRM, SCM, Finance, Inventory and Procurement. The design is cross-functional: each module is described by its own role and by how it contributes to the full enterprise process.

Module	Role in Nishat Textiles	Relevant ERP capability	Integration value
HRM	Employee records, attendance, work schedules, shift visibility, payroll inputs and labour cost analysis.	Supports attendance logs, approvers, payroll contracts, work entries, payslips and payroll analysis.	Connects workforce time to production and payroll.
CRM	Customer records, wholesale/export leads, retail relationships, quotations and sales pipeline visibility.	Supports leads, opportunities, pipeline analysis, expected revenue and quotation workflows.	Connects demand signals to inventory and production planning.
SCM / MRP	Production planning, BoMs, work centres, manufacturing orders and production scheduling.	Supports bills of materials, multilevel BoMs, work centres, dependencies and production analysis.	Aligns manufacturing activity with demand and material availability.
Finance	Invoices, vendor bills, ledgers, analytic accounting, costing and financial reporting.	Supports customer invoices, vendor bills, multi-currency activity, analytic accounting and year-end closing.	Turns operational transactions into auditable financial information.
Inventory	Single source of truth for stock across mills, warehouses, retail stores and e-commerce channels.	Supports stock moves, advanced routes, lead times, replenishment and barcode-enabled operations.	Directly reduces overselling, stockouts and manual reconciliation.
Procurement	Supplier management, purchase orders, tenders, material replenishment and vendor performance.	Supports quotations, purchase agreements, purchase orders, reordering rules and purchase reporting.	Connects supplier decisions to production and stock levels.

The value of ERP comes from the connections between modules. A customer order in CRM/Sales should reserve stock in Inventory, create a picking task in the warehouse, update Finance through invoicing, and influence SCM/MRP if replenishment or production is needed. Procurement then acts on supplier-side requirements, while HRM helps quantify labour capacity and payroll effects. This end-to-end linkage is what differentiates ERP from a collection of unrelated applications.

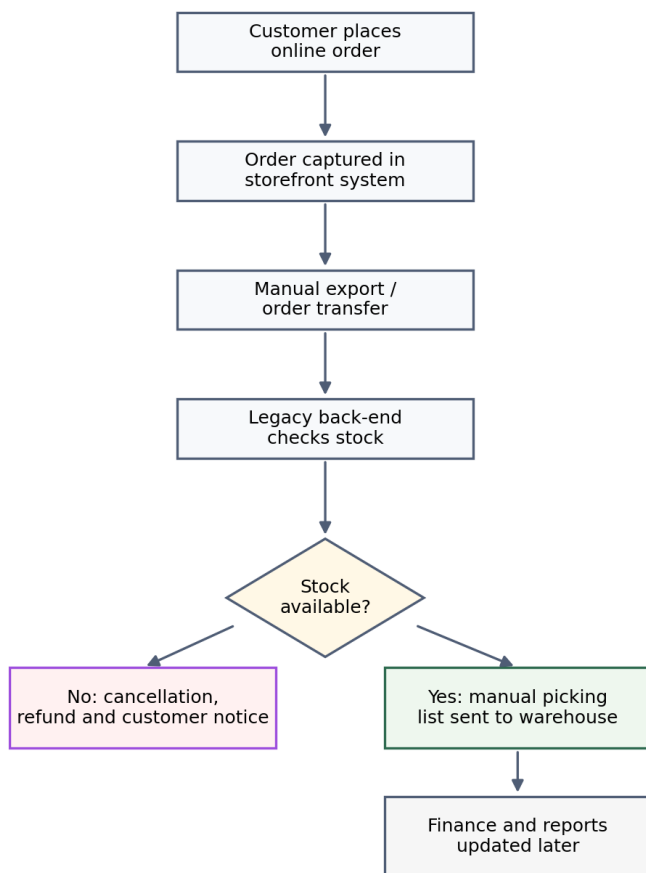
Section 5 – Business Process Mapping

Business process mapping is used to compare the current manual workflow with the proposed integrated ERP workflow. This is necessary because the problem is not only old software; it is an inefficient sequence of activities. Good process redesign begins by understanding the existing workflow before proposing the target model (Bahramnejad, Sharafi and Nabiollahi, 2015).

5.1 As-Is process

In the As-Is process, an online order is captured separately from the legacy warehouse system. Staff then transfer order details manually or through delayed exports. The warehouse checks stock after the order has already been placed, creating a risk that the product is unavailable. If stock is unavailable, staff must cancel the order, process a refund and communicate with the customer. If stock exists, fulfilment proceeds through a manual picking list and finance reporting is updated later.

As-Is Process: Manual E-Commerce Fulfilment



Main weaknesses: delayed inventory visibility, duplicate data entry, stockout risk and slow reporting.

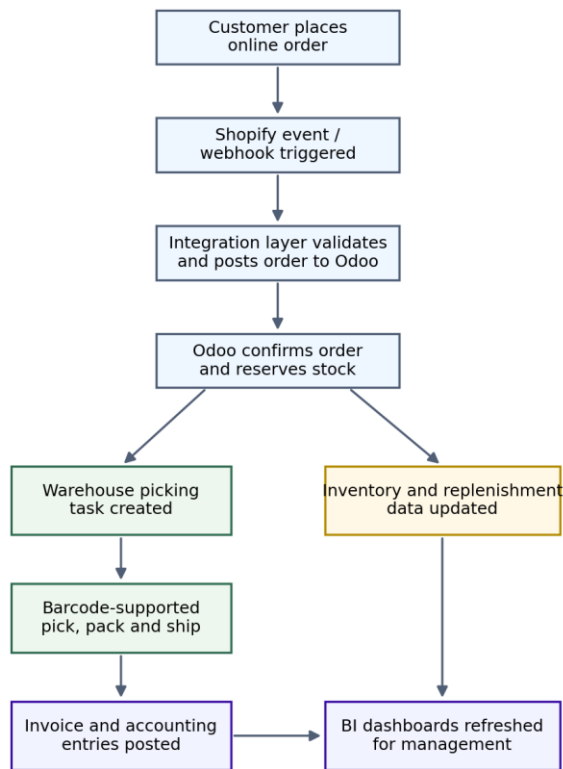
Figure 1: As-Is e-commerce order fulfilment process

Figure 1 shows why the current process is fragile. It separates customer demand from stock reservation and places manual handling between order capture and fulfilment. The decision point “Stock available?” occurs too late, after the customer has already placed the order.

5.2 To-Be process

In the To-Be process, the customer order triggers an online event or webhook. An integration layer validates the order and posts it into Odoo. Odoo confirms the order, reserves the available stock, generates a warehouse picking task and updates inventory records. Once the order is picked, packed and shipped, invoice and accounting entries are posted and the BI dashboard is refreshed.

To-Be Process: Integrated Odoo Workflow



Main improvements: real-time reservation, integrated accounting, traceable fulfilment and analytics-ready data.

Figure 2: To-Be integrated Odoo order fulfilment process

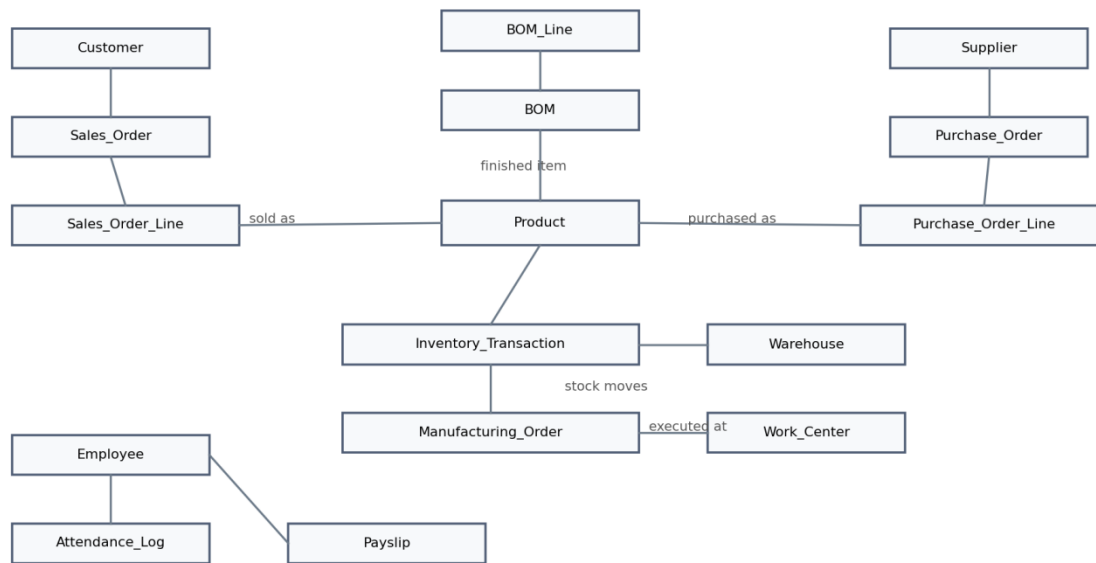
Figure 2 shows the target workflow. The main improvement is not just speed; it is control. Stock reservation happens before fulfilment, inventory and finance are updated through the same workflow, and management dashboards receive cleaner transaction data.

Process dimension	As-Is state	To-Be state
Order capture	Separate storefront record	ERP transaction created through integration
Inventory visibility	Delayed manual check	Reservation-based stock control
Fulfilment trigger	Manual picking list	Workflow-generated picking task
Finance update	Updated after manual reconciliation	Invoice and accounting entries linked to the order
Reporting	Retrospective, spreadsheet-based	Near-real-time dashboard data
Error risk	High duplicate-entry and mismatch risk	Reduced through shared master data and validation

Section 6 – Database Design

The database design is presented as a logical model rather than a copy of Odoo's internal physical schema. The aim is to show which business entities must be consistently related for the ERP to support integrated operations and analytics. The design focuses on master data, transaction data and operational execution data.

Logical ERP Database Design (ERD Summary)



The model shows the shared master-data and transaction relationships needed for integrated reporting.

Figure 3: Logical ERP entity relationship design

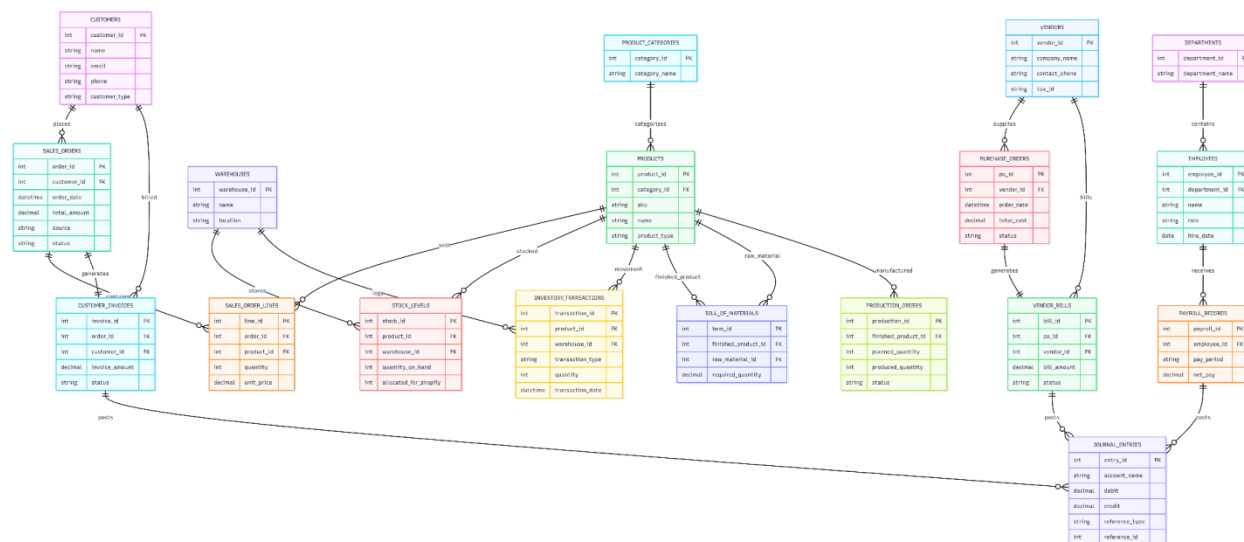
Figure 3 shows how customer, supplier, product, warehouse, manufacturing and HR records connect. The Product table sits at the centre because it appears in sales orders, purchase orders, inventory movements, bills of materials and manufacturing orders. This central relationship is critical for accurate costing, stock visibility and dashboard reporting.

Core table	Primary purpose	Key fields
Customer	Stores retail, wholesale and online customer records.	Customer_ID, Name, Channel, Segment, Region
Product	Maintains item master data across manufacturing and retail.	Product_ID, SKU, Product_Type, Unit_Cost, Sales_Price
Warehouse	Identifies stock locations such as mills, distribution centres and stores.	Warehouse_ID, Location, Warehouse_Type
Inventory_Transaction	Records all stock movements and adjustments.	Txn_ID, Product_ID, Warehouse_ID, Txn_Type, Quantity, Txn_Time
Sales_Order	Holds online, retail and wholesale order headers.	SO_ID, Customer_ID, Order_Date, Channel, Status
Sales_Order_Line	Captures product-level order details.	SO_Line_ID, SO_ID, Product_ID, Quantity, Unit_Price
Supplier	Stores vendor master records.	Supplier_ID, Name, Category, Payment_Terms
Purchase_Order	Records supplier-side purchase transactions.	PO_ID, Supplier_ID, Order_Date, ETA, Status
BOM / BOM_Line	Defines finished product structures and component requirements.	BOM_ID, Finished_Product_ID, Component_Product_ID, Qty_Per_Unit
Manufacturing_Order	Tracks production execution against plan.	MO_ID, Product_ID, BOM_ID, WorkCenter_ID, Planned_Qty, Status
Employee / Attendance / Payslip	Links labour presence and payroll to operational periods.	Employee_ID, Attendance_ID, Check_In, Check_Out, Payslip_ID, Net_Pay

The database model supports the BI (Business Intelligence) layer because it creates reliable relationships between the data used for KPIs (Key Performance Indicator). Sales orders and order lines support demand analysis. Inventory transactions support stock accuracy and replenishment. Purchase orders support supplier performance. Manufacturing orders support schedule adherence and OEE analysis. HR and payroll records support labour productivity and cost analysis. Without clean relationships across these tables, dashboards would reproduce the same inconsistency found in the current system.

The logical designed ERD for the Nishat Textiles under the assumptions we have stated earlier

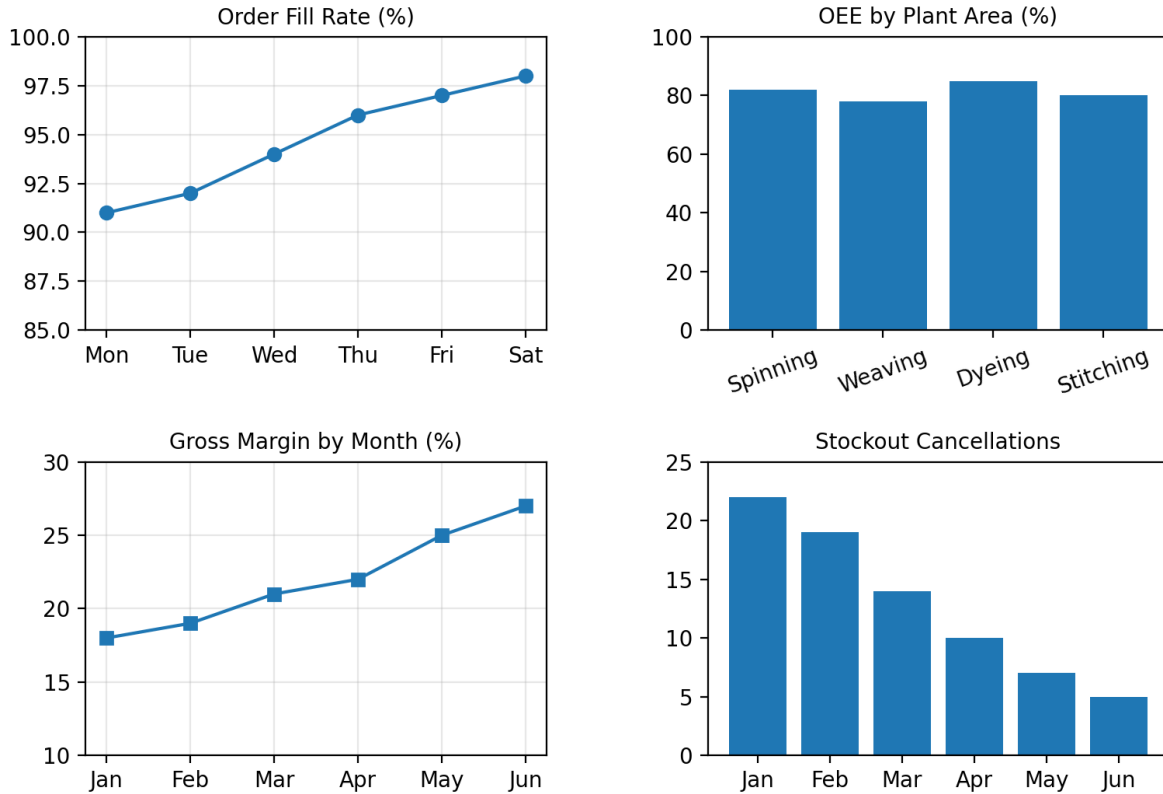
Note: The ERD doesn't follow the database of the Odoo ERP System



Section 7 – Analytics & BI Dashboard

The BI dashboard should be designed around managerial decisions rather than chart volume. ERP analytics are useful when they help managers identify exceptions, understand causes and act before losses increase. The proposed dashboard combines sales, inventory, procurement, manufacturing, finance and HR indicators.

Illustrative ERP Business Intelligence Dashboard



Illustrative values only; actual dashboard baselines must be taken from live ERP transactions.

Figure 4: Illustrative ERP business intelligence dashboard

Figure 4 is an illustrative dashboard, not an audited performance chart. It demonstrates the type of visual decision support the ERP should provide: order fill rate, plant efficiency, gross margin and stockout cancellations. The values shown are sample design values and must be replaced with live ERP data during implementation.

KPI	Calculation	Main data source	Decision value
Order fill rate	Fulfilled orders ÷ total orders	Sales orders and inventory	Shows whether demand is being served without avoidable cancellations.
Stockout cancellation rate	Orders cancelled for stockout ÷ total online orders	Sales orders and cancellation reasons	Measures revenue leakage caused by inventory mismatch.
Inventory turnover	Cost of goods sold ÷ average inventory	Inventory and finance	Shows whether stock is productive or tying up cash.
Supplier lead-time adherence	On-time receipts ÷ scheduled receipts	Purchase orders and receipts	Evaluates supplier reliability.
Production schedule adherence	Manufacturing orders completed on time ÷ total manufacturing orders	Manufacturing orders	Measures planning discipline and execution reliability.
OEE	Availability × Performance × Quality	Work centre and production records	Identifies equipment losses and operational bottlenecks.
Gross margin by product line	Revenue – direct product cost	Sales and accounting	Supports product mix, pricing and collection decisions.
Attendance variance	Scheduled hours – actual hours	HRM and attendance records	Shows labour discipline and capacity gaps.

The KPI framework is cross-functional. Stockout cancellation rate is a sales metric, but it also affects procurement and production planning. OEE is a manufacturing metric, but it influences lead times, available stock and customer promise dates. Gross margin is a finance metric, but it should inform pricing, design decisions and replenishment priorities. This cross-functional interpretation is what makes the dashboard an ERP dashboard rather than a set of isolated departmental reports.

Section 8 – ERP Implementation Lifecycle

A phased implementation is recommended. A single big bang migration would expose the company to high operational risk because manufacturing, warehousing, online sales and finance must continue while the system is changing. ERP implementation research emphasizes the importance of executive support, clear scope, process readiness, training, user involvement and disciplined change management (Umble, Haft and Umble, 2003; Shad, Chen and Azeem, 2012).

Lifecycle phase	Main activities	Key deliverables
Planning	Create project charter, steering committee, scope boundaries, budget and success measures.	Approved governance structure and project roadmap.
Requirement analysis	Map As-Is processes, identify pain points, audit master data and define To-Be requirements.	Functional requirements, process maps and data-cleansing plan.
Customization and configuration	Configure Odoo modules, approval workflows, warehouse routes, reordering rules and integration layer.	Configured test environment and integration specification.
Testing	Perform unit testing, integration testing, user acceptance testing, stress testing and mock cutover.	Test evidence, reconciliation results and UAT sign-off.
Training	Deliver role-based training for warehouse, finance, HR, procurement, sales and management users.	User guides, training completion records and super-user network.
Deployment	Migrate clean data, activate modules in phases, monitor issues and provide hypercare support.	Stable go-live, issue log and post-go-live support plan.

8.1 Recommended rollout sequence

1. Phase 1: Sales, Inventory and Procurement. This phase addresses the most visible operational pain points: online stock visibility, order reservation, warehouse picking and material replenishment.
2. Phase 2: Manufacturing and SCM. Once stock and order data are reliable, production planning, BoMs, work centres and manufacturing orders should be configured for the mills.
3. Phase 3: Finance and HRM. Finance and HR should be migrated with strong reconciliation controls because payroll, ledgers and cost reports are sensitive areas.
4. Phase 4: Advanced BI and continuous improvement. After stable operations, dashboards can be expanded to forecasting, supplier analytics, margin analysis and workforce productivity.

The phased approach is not slower for the sake of caution; it is a control strategy. It gives the company time to test each process, train users, stabilize master data and prevent the new ERP from becoming a digital copy of old inefficiencies.

Section 9 – Challenges & Risks

ERP projects often fail because organizations underestimate process, data and people risks. In this case, the technology shift from a legacy database to a modern integrated ERP is only one part of the challenge. The company must also change how departments coordinate work.

Risk	Why it matters	Likely impact	Mitigation
Resistance to change	Users may trust old workarounds and spreadsheets more than the new ERP.	Low adoption, shadow systems and incomplete data.	Executive sponsorship, super users, role-based training and visible quick wins.
Data migration failure	Legacy records may contain duplicates, obsolete SKUs or inconsistent customer and supplier data.	Wrong inventory, inaccurate reports and broken workflows.	Data cleansing, migration rehearsals, validation rules and reconciliation checks.
Integration breakdown	Shopify–Odoo synchronization depends on APIs, middleware and exception handling.	Delayed orders, overselling and customer refunds.	API monitoring, retry logic, error queues and integration ownership.
Scope creep	Departments may request extra custom features during implementation.	Budget overrun, delayed go-live and unstable configuration.	Strict change control and phased priorities.
Training weakness	Factory, warehouse, finance and HR users need different learning paths.	User errors and process non-compliance.	Function-specific training, practical simulations and competency sign-off.
Security and control risk	New API integrations and cloud access increase permission-management requirements.	Unauthorized access or weak auditability.	Role-based access control, audit logs, secure API credentials and periodic access reviews.

Two risks require special emphasis. First, data quality must be addressed before go live. If product, supplier, customer and warehouse records remain inconsistent, the new ERP will simply produce faster inaccurate reports. Second, users

must be discouraged from maintaining parallel spreadsheets as the “real” source of truth. If shadow systems continue, the ERP will be technically live but operationally weak.

Section 10 – Final Recommendations

Recommendation	Rationale	Expected improvement
Start with master data governance	Clean product, customer, supplier and warehouse data before migration.	More accurate stock, finance and dashboard outputs.
Implement sales inventory procurement first	These functions directly address overselling, stockouts and manual re entry.	Early operational benefits and stronger user confidence.
Treat Shopify integration as a monitored connector project	APIs and webhooks require validation, retry logic and exception handling.	Fewer order failures and better auditability.
Use phased implementation rather than big bang	Manufacturing and retail operations must continue during migration.	Lower go-live risk and better learning between phases.
Design dashboards around decisions	Each KPI should answer a management question.	More useful analytics and less visual clutter.
Invest in training and change champions	ERP adoption depends on user confidence and local ownership.	Less resistance and higher process compliance.
Run parallel reporting before full cutover	Old and new figures must be reconciled before management relies fully on ERP reports.	Safer financial transition and stronger trust in the system.

The organization should measure ERP success by process outcomes, not by installation alone. A successful implementation means that one customer order can move from online capture to stock reservation, warehouse picking, invoicing and management reporting without manual re-entry or avoidable data conflict. The ERP should create a single operating truth across manufacturing, retail, supply chain, finance and HR.

Conclusion

The case presents a clear ERP problem: a vertically integrated textile and retail organization is constrained by a fragmented legacy architecture. The current environment creates duplicate data entry, unreliable inventory visibility, delayed reports, weak communication and operational risk. Odoo is a defensible ERP choice because its modular applications support the required functions and because its integration capabilities can connect online sales activity with internal ERP workflows. However, the report also shows that software selection alone will not guarantee success. The critical success factors are process redesign, master-data quality, integration governance, testing, training and executive ownership.

If implemented with discipline, the proposed ERP design should improve stock accuracy, reduce order cancellations, shorten reporting cycles, support better production planning and provide management with analytics-driven decision support. These benefits should be verified after deployment through baseline KPIs such as order fill rate, stockout cancellation rate, reporting cycle time, production schedule adherence and inventory turnover. This makes the proposal practical, measurable and aligned with the requirements of an academic ERP solution design project.

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Appendix A – Requirement Compliance Checklist

Requirement	Where addressed
Company introduction	Covered in Section 1 with background, industry type, systems and assumptions.
Problems in current system	Covered in Section 2 with duplicate data, inventory issues, integration, reporting and communication problems.
ERP solution design	Covered in Section 3 with ERP selection, justification and expected improvements.
ERP modules	Covered in Section 4 with HRM, CRM, SCM, Finance, Inventory and Procurement.
Business process mapping	Covered in Section 5 with As-Is and To-Be workflows and diagrams.
Database design	Covered in Section 6 with logical ERD and table relationships.
Analytics and BI dashboard	Covered in Section 7 with KPIs, dashboard design and chart explanation.
Implementation lifecycle	Covered in Section 8 with planning, requirements, customization, testing, training and deployment.
Challenges and risks	Covered in Section 9 with mitigation actions.
Final recommendations	Covered in Section 10 and conclusion.